



THE UNIVERSITY OF
AUCKLAND
Te Whare Wānanga o Tāmaki Makaurau
NEW ZEALAND

AGILE AND LEAN SOFTWARE DEVELOPMENT

SOFTENG 761



COURSE OVERVIEW

Teaching Team

Lecturers

▶ [Dr. Paramvir Singh](#)

- ▶ Module 1 (Project Rationale) and Module 2 (Understanding Agile)
- ▶ p.singh@auckland.ac.nz
- ▶ B303 Room 423

▶ [Dr. Reza Shahamiri](#)

- ▶ Module 3 (Project)
- ▶ reza.shahamiri@auckland.ac.nz
- ▶ B405 Room 669
- ▶ GitHub ID: rshahamiriua

Teaching Team

Graduate Teaching Assistants

▶ Hoda Yamani

▶ hyam650@aucklanduni.ac.nz

▶ GitHub ID: h-yamani

▶ Rabia Rao

▶ rrao583@aucklanduni.ac.nz

▶ GitHub ID: Rabia24-sudo

How to Contact

Ways to Contact

- ▶ Email (Subject: SOFTENG761)
- ▶ **Ed Discussion** – mainly for student-to-student communication – teaching team will reply in 48 hours if any questions are unanswered
- ▶ Virtual meeting (on demand)
 - ▶ Email in advance for a suitable time (outside regular lectures)
- ▶ Open Office: on-demand
- ▶ Extra support sessions for overseas students (if any) can be scheduled on demand

How to Contact Queries

- ▶ All Module 1 and 2 related questions to Paramvir
- ▶ All project-related questions to be directed to Reza or GTAs
- ▶ All mark queries are via email
- ▶ Other generic questions and concerns to either Reza or Paramvir

How to Contact

For project-related questions, please ensure that you read the project document on Canvas and check the FAQs at the end of the document before posting a new question.

Lectures Timing

Lectures

- Monday 2 – 4 PM (423-348)
- Wednesday:
 - ▶ Weeks 1 to 3: 10 AM -12 PM at 405-422
 - ▶ Weeks 4 to 11: 1:00 PM - 3:00 PM at 423-348

Labs

- Tuesday 10 AM to 12 PM (401-307 , 401-311, 401-312)
- Wednesday 10 AM to 12 PM (except weeks 1-3)(401-307 , 401-311, 401-312)



Reza Shahamiri (he/him)

- Senior Lecturer in Software Engineering and AI @ the UoA
- BE and ME in Software Engineering, PhD in Computer Science
- Deep Learning Software Engineer
- Research Interests:
 - Development of novel, intelligent health care systems using deep learning technologies (impaired speech, dementia, autism, depression)
 - Software Test Oracles
- List of my available PhD projects [here](#)
- My Team: DeepNet Discovery Network ([2DN](#))
- Professional Android Developer ([SEMA](#))



Paramvir Singh (he/him)

- Professional Teaching Fellow (Software Engineering) in School of Computer Science
- BTech and ME in Computer Science and Engineering, PhD in Software Engineering
- Teaching and Research Interests:
 - Software design and architecture (COMPSCI 230, COMPSCI 331, ENGGEN 131)
 - CS and SE education
 - Recent project – Risks in team projects
 - Note taking and Journal writing
 - Academic songwriting (YT page - [@ParamvirEducationalSongs](#))
 - Programming songs



Hoda Yamani– Graduate Teaching Assistant

- ▶ PhD candidate in Computer Systems Engineering
- ▶ BE and ME in Software Engineering
- ▶ Machine Learning Engineer and Research Assistant in robotics and AI
- ▶ Research Interest
 - ▶ Reinforcement learning
 - ▶ Deep learning and intelligent systems
 - ▶ Robotic autonomy
- ▶ PhD Project
 - ▶ Developing memory-guided and biologically inspired methods for sample-efficient reinforcement learning



Rabia Rao

- PHD student in Computer Systems Engineering at the UOA
- BE(Hons) and ME in Computer Engineering
- Research interest
 - Deep learning, predictive modelling, and AI-driven healthcare solutions, particularly in healthcare and psychology

Students

- ▶ BE(Hons) Part 4
 - ▶ Computer Systems Engineering
 - ▶ Software Engineering
 - ▶ Others?
- ▶ Master of Engineering Studies
 - ▶ SE, CSE, Others?
 - ▶ Masters Info Tech
- ▶ Agile Experience?

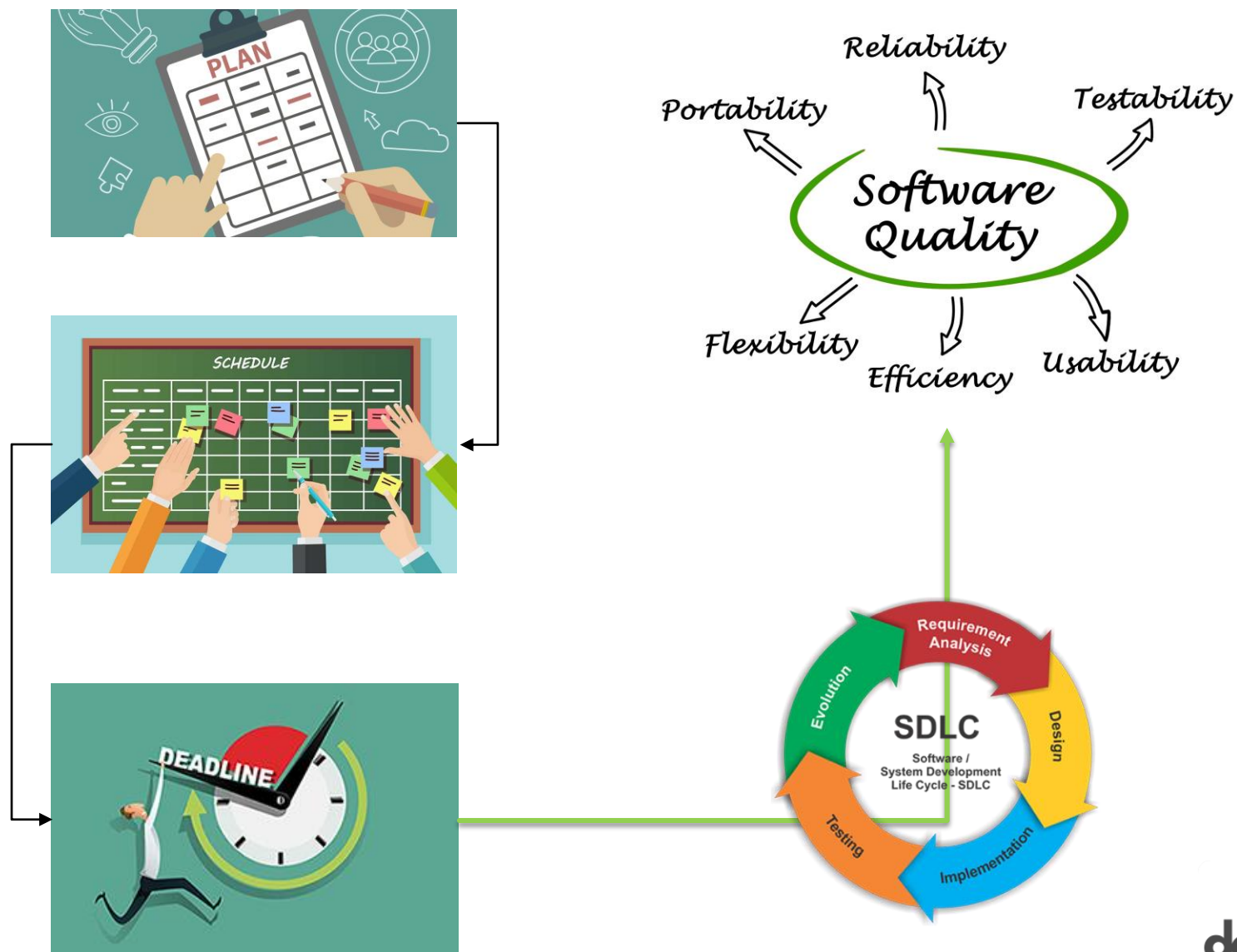
Class/Part 4 Representatives

Pre-Requisites*

- ▶ For BE(Hons) Part 4 students:
 - ▶ SOFTENG 306
 - ▶ COMPSYS 302
- ▶ For Post Graduate students:
 - ▶ Strong OO design and Programming skills
 - ▶ Git/GitHub experience
- ▶ For Everyone:
 - ▶ Good teamwork skills and a fun attitude to learning
 - ▶ Ability to commit to weekly deliverables

* If unsure or have any concerns, please double check with the lecturers immediately (in the first week) in case you would like to withdraw.

What is this course?



What is this course?

- ▶ **Agile and Lean software development**
 - ▶ Focus on process and practices, self-organizing teamwork and project management, to deliver high quality software products.
- ▶ **An advanced software engineering course**
 - ▶ Builds your previous skills and experiences from other courses/ industry, apply them well!
- ▶ Has both team and individual components
- ▶ Multiple forms of assessment:
 - ▶ project (practical) deliverables,
 - ▶ essays,
 - ▶ reflection,
 - ▶ **no** final exam

What is this course?

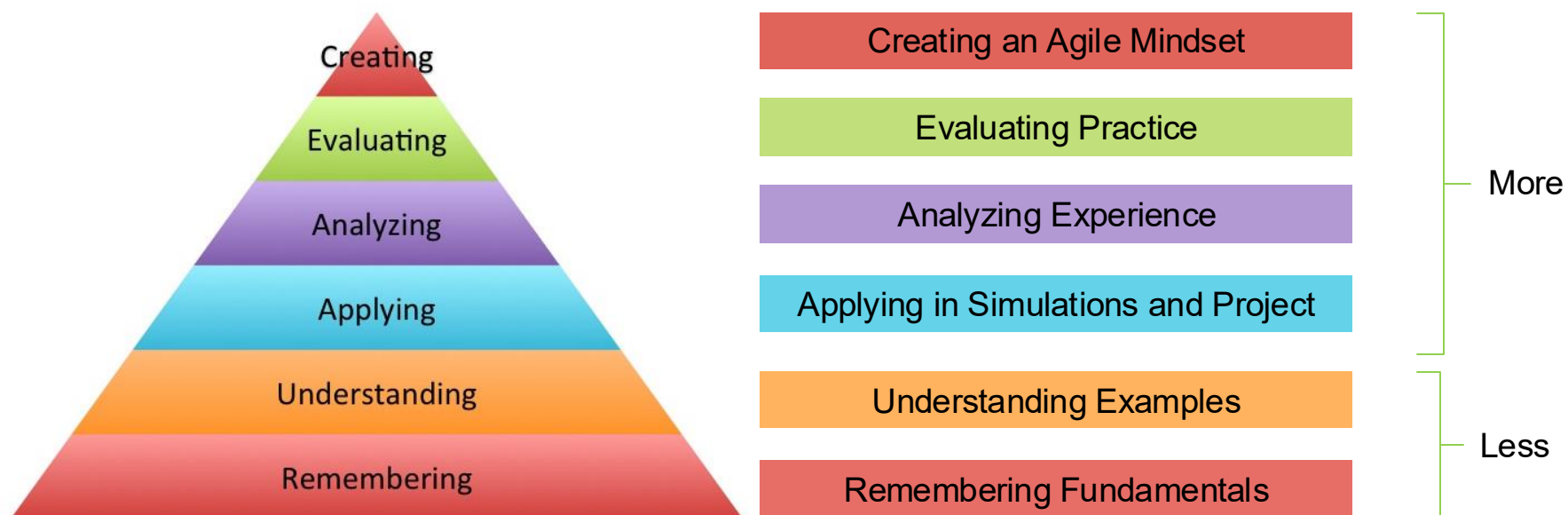
It is **NOT**...

- ▶ **Not** a course to teach any programming language or technology.
 - ▶ As Part IV and PG students, it is assumed you can pick up any language/technology easily.
- ▶ **Not** a 100% lecture-based not a 100% project-based course.
 - ▶ Contains multiple forms of learning through lectures, flipped-classroom, hands-on projects, classroom presentations, individual research, and writing.
- ▶ **Not** a course to formally count towards professional work hours requirements.

Learning Outcomes

- ▶ **Understand agile and lean software development** fundamental knowledge and **theory**
- ▶ **Apply** theoretical knowledge of **agile and lean concepts** to a **real software project**
- ▶ Demonstrate **effective communication** with **stakeholders** to comprehend and **clarify requirements**, communicate technical barriers, resolve differences, develop a **project plan**, and **deliver** a software product
- ▶ Demonstrate **good project planning** and **management practices**
- ▶ Critically **reflect on practical project experience**

Course Design



Based on Bloom's Taxonomy of Learning (Revised, 2001)

Course Structure

- ▶ **Module 1 & 2: Project Rationale & Understanding Agile** (handled by Paramvir)
 - ▶ Fundamental knowledge (Week 1-3)
 - ▶ Assessed via three written assessments
 - ▶ Overall, 20% of the course weight
- ▶ **Module 3: Application via a project** (handled by Reza)
 - ▶ Practical team project (Week 4-11)
 - ▶ 80% of the course weight

Course Structure

Module 1: Project Rationale

Module 2: Understanding Agile

- ▶ Individual work
- ▶ Both modules run in parallel
- ▶ Module 1: Project Rationale
 - ▶ Justifying projects, broadly and deeply.
 - ▶ Preparing for our project *before* meeting the client.
 - ▶ Two deliverables (A1 and A3)
- ▶ Module 2: Understanding Agile
 - ▶ Understanding the implications of Agile, what will work, what won't?
 - ▶ One deliverable (A2)

Weeks 1-3: Individual submissions worth 20%

Course Structure

Module 3: Project

- ▶ Teams of eight students
- ▶ Each team has a separate project
 - ▶ Ensure the team members have **complementary skills**
 - ▶ Think about the main project you're bidding on, and decide what skills the team needs
 - ▶ **Chances are you will be assigned to another project.** Be prepared.
 - ▶ When forming teams, be prepared that members may change their decision
 - ▶ Also, consider changes to the enrolments during the first two weeks
- ▶ The projects are real
 - ▶ Real clients and real users
- ▶ Each team has a client (aka Product Owner - PO)
- ▶ Software Methodology: Scrum
- ▶ Objective: To develop the software PO needs, following Scrum
- ▶ Three **sets of** deliverables

Weeks 4-11: Team project worth 80%

Course Structure

Module 3: Project

- ▶ We have multiple projects in which you can submit your EoI
- ▶ You need to form up your team this week
- ▶ Appoint a Scrum Master (not a project manager) – this week
- ▶ Submit your EoI and Team Portfolios by end of Week 2
- ▶ Your project allocation will be finalized early Week 3
- ▶ Scrum Master to contact the PO and team to start the project immediately

Course Structure

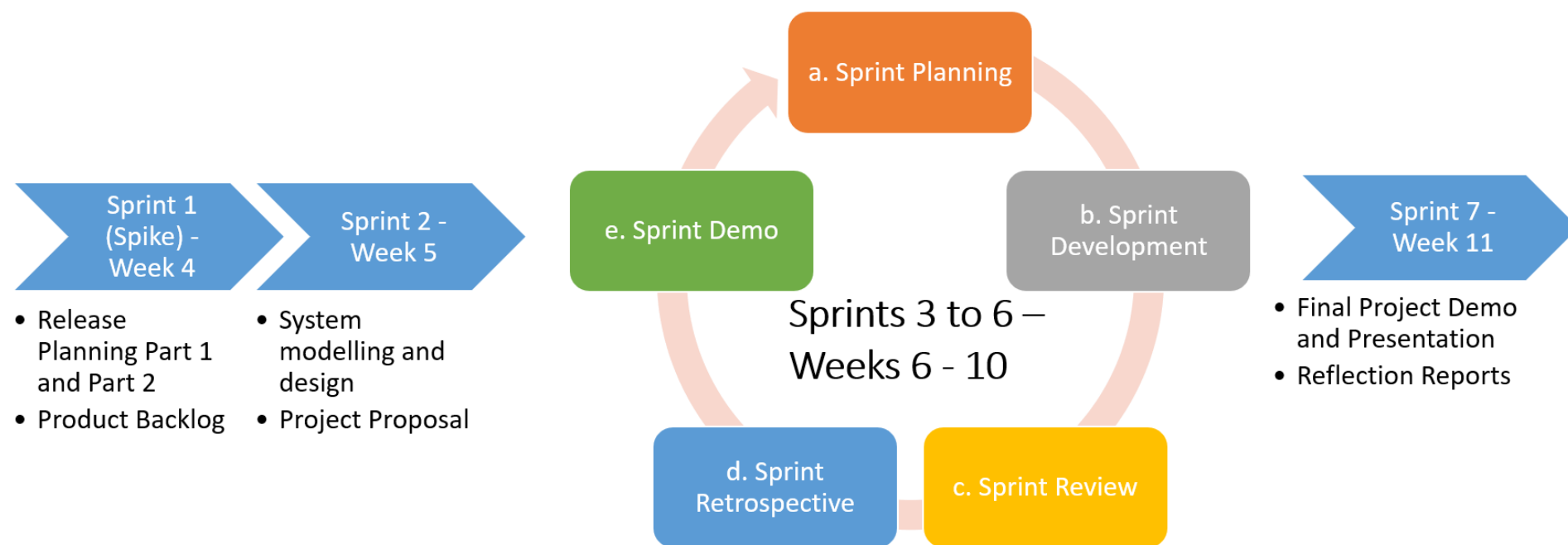
Module 3: Project

- ▶ You need to meet/e-meet with your PO every week
 - ▶ For overseas/non-Auckland-based POs or students use Zoom, Teams, etc. and record the sessions for those who can't attend
- ▶ Meeting locations: you can use the [labs](#) for your meetings, or request to book meeting rooms:
 - ▶ booking in advance is required for meeting rooms (ask GTAs for help)

Course Structure

Module 3: Project Process

- ▶ 7 Sprints
- ▶ Each sprint is one week



Course Structure

Module 3: Project

Sprint Demos

- ▶ Happens during Sprints 3 to 6
- ▶ Demo sprint plan of the current sprint and implementations - in class
- ▶ Set Spring start weekday to the day after demos
- ▶ Each demo is 5 minutes
- ▶ Delivered by two-three students – **all students** must present in at least one sprint demo

Course Structure

Module 3: Project

Sprint 7

- ▶ Happens on Week 11 – last week of the project
- ▶ Closure of the project
- ▶ Final demo and presentation
- ▶ Final deliverable submissions

Please be advised that your Part IV project final report is usually due during this sprint; hence it is very important that you follow the process properly and do not use Sprint 7 as a full development sprint. You may only have time to make some minor changes before you submit the final deliverables.

Course Structure

Module 3: Project

Deliverables

- ▶ The tasks you perform in this project are **holistically** assessed via three sets of deliverables (and the four Sprint Demos) that total to 80% of the course mark, and a Team Portfolio that is marked as part of Project Proposal but needs to be submitted in advance.
- ▶ We will consider the entire agile process, the quality of the product you produced, whether you applied best software engineering principles and practices, your team working skills, your negotiation skills, the feature richness and complexity of your product, etc.
- ▶ Although each deliverable has its own weight, your performance in one deliverable will still affect the others as part of our holistic assessment and based on **your individual contributions**.

Course Structure

Module 3: Project Deliverables

- ▶ Team Portfolios
- ▶ Project Proposal (includes portfolios again) – 15 marks
- ▶ Project Demos and Presentations (35 Mark) includes:
 - a) Sprint Demos (8 marks overall)
 - b) Final Demo and Presentation (12 marks, where 10 marks to be allocated individually) – due week 11
 - ▶ Individual Presentation Mark will be based on your performance in both “Sprint Demos” and “Final Demo and Presentation”.
 - ▶ **Attendance is mandatory for all students**
 - c) Code Quality and Professionalism (15 marks)
- ▶ Reflection Report (30 individual mark) – due week 11
 - a) Individual Reflection (5 marks)
 - b) Participation Report and Peer-Review (25 marks)
- ▶ Scrum Master’s Member Participation Report

Project Schedule

	To Do	Deliverables
Week 1	Formation and confirmation of teams (consider enrolment changes too)	Teams (on Canvas)
Week 2	Bid for projects, submit the team portfolios with photos, and confirmation of Scrum Masters – this will be sent to your Product Owners	Eols- Friday Team Portfolios – Sunday
Week 3	- Project Allocation (by the teaching team) - Submission of team portfolios to Product Owners (by the teaching team) - Students to sign the Project Agreement form and submit it on Canvas	Project Agreements
Week 4	Sprint 1	Create a GitHub repo and add all team members and GTAs
Week 5	Sprint 2	Project Proposal
Week 6	Sprint 3	Sprint 3 Demo GitHub Repository Information Sheet and Ownership Transfer
Study Break		
Week 7	Sprint 4	Sprint 4 Demo
Week 8	Sprint 5	Sprint 5 Demo
Week 9	System's Week – No Course work	
Week 10	Sprint 6	Sprint 6 Demo Submit all Sprint Demo Slides
Week 11	Sprint 7	Final Project Demo and Presentation (and code submission) Reflection Report Scrum Master's Member Participation Report

Reminders:

1. Do not forget to book your sprints
2. Do not forget to book your Final Project Demos

Check Project Doc/Canvas for the links.

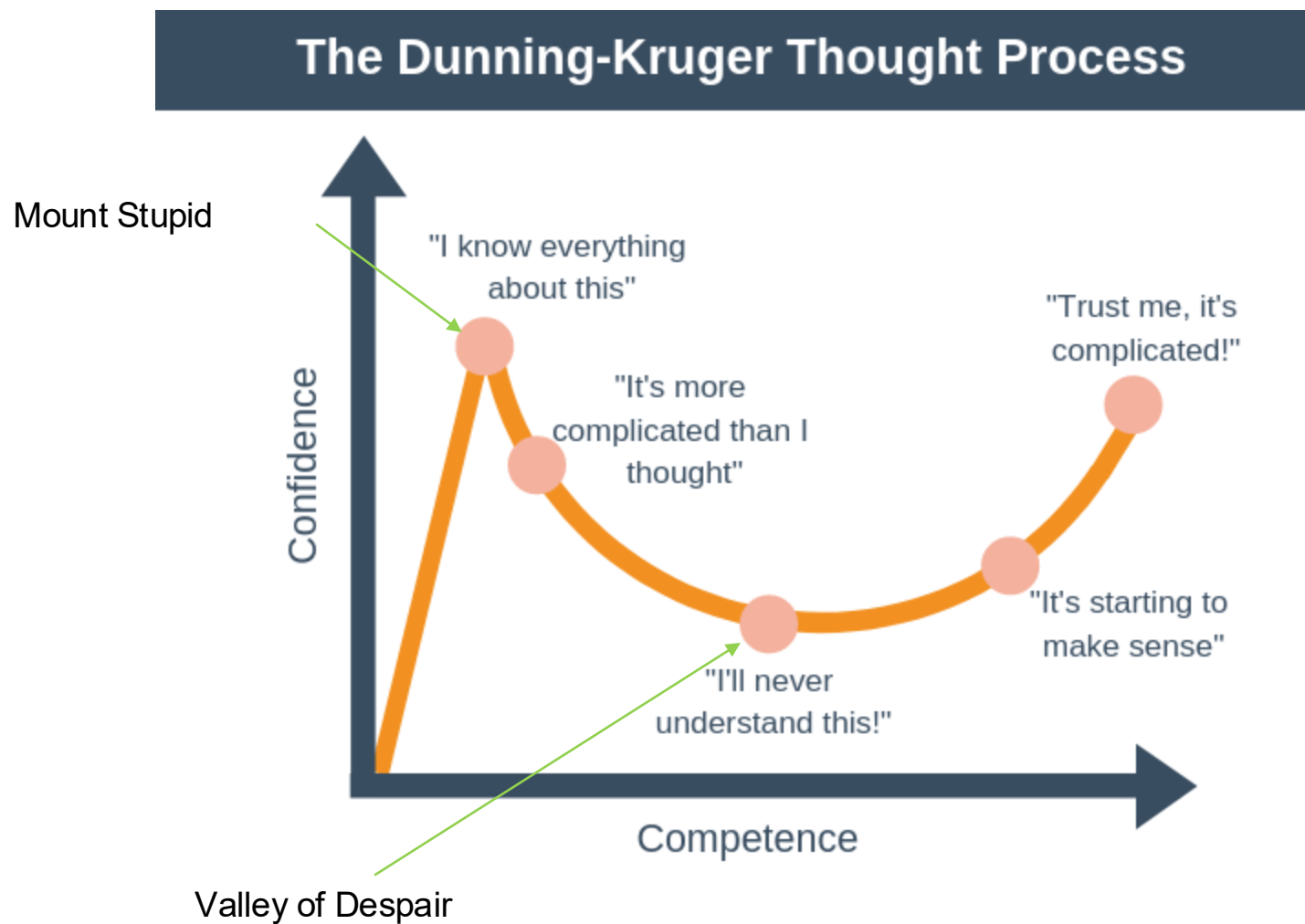
Imposter Syndrome

Video from
Udacity.com

Dunning–Kruger effect

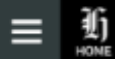


Dunning-Kruger effect



Academic Integrity

- ▶ Why is academic integrity important?
 - Student Academic Conduct Statute (2020): “Such integrity maintains **the reputation and quality** of its qualifications and protects their international recognition.”
 - Engineering New Zealand’s Code of Ethical Conduct: “In performing, or in connection with, your engineering activities you must act with honesty, objectivity, and integrity ...”
 - Academic dishonesty has long-term consequences.


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NEW ZEALAND | Education

University of Auckland cheating claims: 'Prolific' abuse of online exams by students on group calls

15 Jun, 2021 | 08:06 AM

3 minutes to read

The New York Times

Volkswagen Engineer Gets Prison in Diesel Cheating Case

f s t v p l



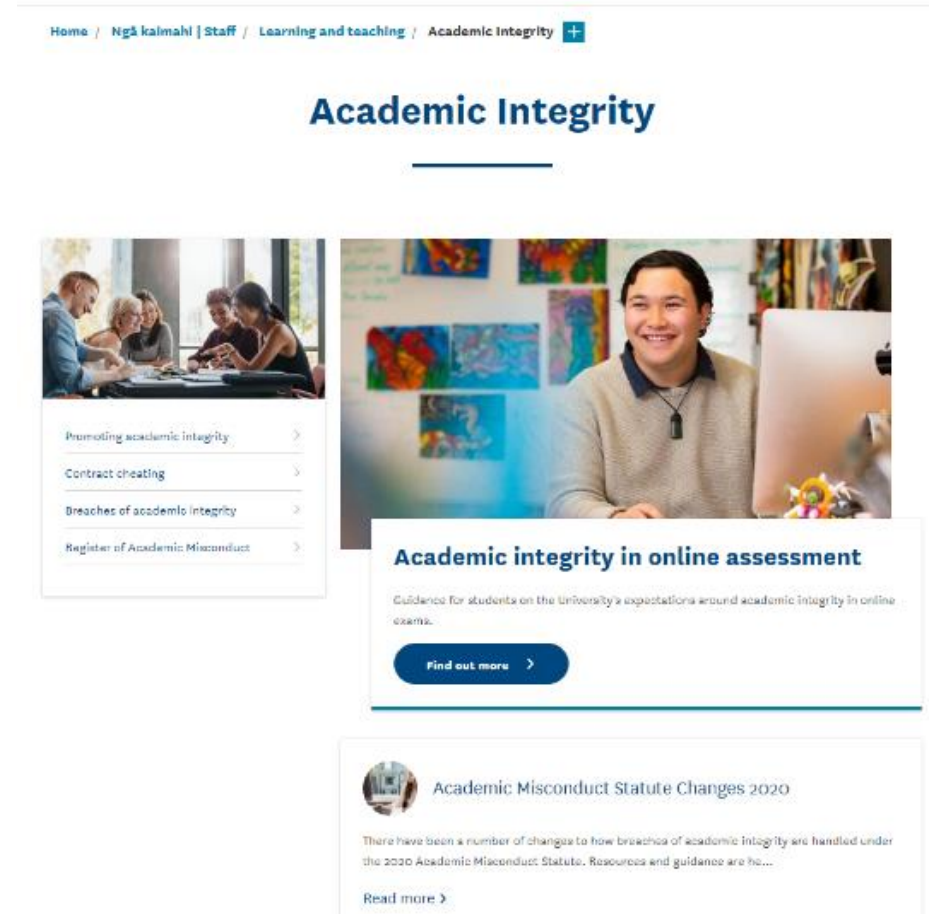
James Liang, a Volkswagen engineer, is the first employee sent to prison in the scandal that has tainted the company's reputation and cost it billions in fines and settlements with consumers. Virginia Luciano/Detroit News, via Associated Press

By Bill Vlasic
Aug. 29, 2017

Discovery Network

Academic Integrity at UoA and FoE

- 120 misconduct cases (coursework and exam, incl. plagiarism, collusion, contract cheating, etc.) in the Faculty of Engineering in 2020.
- The cases are recorded in the Register of Academic Misconduct.
- Information about Academic Integrity at the University of Auckland can be found here:
<https://www.auckland.ac.nz/en/staff/learning-and-teaching/academic-integrity.html>



Generative AI in Modules 1 and 2

Module 1 and 2 (Pre-Project Research)

- implements the [Two-Lane Approach to assessment](#), as Lane 2: Uncontrolled assessment, that is, the use of AI to assist students.
- You are allowed to use generative AI (such as ChatGPT) for A1, A2, and A3. But you should refer to the specific instructions of each assignment for more details. More specifically:
 - A1: It is highly encouraged and expected you will use ChatGPT, and even need to elaborate on using it.
 - A2 (*partially controlled*): This is a reflection, and therefore you are NOT expected to be using generative AI here.
 - A3: It might be useful to use ChatGPT for some components here, but there's also a personal reflection component.

The usage of any generative AI tools **must be** disclosed in the assignments' deliverables. Failure to disclose may result in the breach of the Statute.

Generative AI in Module 3

- Module 3 implements the [Two-Lane Approach to assessment](#), as Lane 2: Uncontrolled assessment, that is, the use of AI to assist students.

Your use of AI is governed by the following three pillars:

1. If GenAI contributed significantly to a code block or a section of your report, you must disclose it. **Failure to disclose the use of these tools is considered academic dishonesty and will be investigated as plagiarism.**

Generative AI in Module 3

2. Every student is 100% responsible for every line of code and every paragraph in the report, regardless of which student is the original author. During your presentation and Q&A, if you cannot explain any materials or code blocks — even if "inspired" by AI or developed by another team member — you will be graded as if you do not understand the material. A "black box" excuse (e.g., "The AI suggested this") is not an acceptable engineering justification.

Generative AI in Module 3

3. Using AI to generate a complete code, or copying a pre-existing code from a prompt, is prohibited; the project will receive a zero mark. Any report or app that is found to be primarily "AI-authored" (where the student acted mostly as a copy-paster) will receive a mark of zero and may result in formal disciplinary action.

